ANALYSIS REPORT

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `Wave 4.2` quantile/probabilistic campaign into the official `TE Curve Verification Pipeline` offline verification matrix.

Decision

- `Wave 4.2` quantile/probabilistic regression is verified as an exploratory dispersion-aware baseline.
- No quantile/probabilistic candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The project continues to maintain three parallel best surfaces: `Fw`, `Bw`, and `global`.
- Within the quantile/probabilistic package, the strongest forward candidate is `track2h_gaussian_nll_global`.
- Within the quantile/probabilistic package, the strongest backward candidate is `track2h_quantile_p10_p50_p90_Bw`.
- Within the quantile/probabilistic package, the strongest global candidate is `track2h_gaussian_nll_global`.
- The probabilistic package improves the previous `Wave 4.1` robust-loss package on the best `global` and `Bw` surfaces, but not on the best `Fw` surface.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix: `doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary: `output/validation_checks/track2_reference_comparison/2026-06-12-14-44-45__track2_full_directional_family_matrix_track2h_quantile_probabilistic_track2_refresh_2026_06_12/validation_summary.yaml` ;
- per-condition metrics: `output/validation_checks/track2_reference_comparison/2026-06-12-14-44-45__track2_full_directional_family_matrix_track2h_quantile_probabilistic_track2_refresh_2026_06_12/per_condition_metrics.csv` ;
- best-model collage report: `doc/reports/analysis/track2/best_model_collage_report/[2026-06-12]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report: `doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-12]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs: `output/validation_checks/track2_operator_launch_logs/2026-06-12-14-44-41_track2h_quantile_probabilistic_track2_refresh_2026_06_12/` .

Candidate Refresh

The refresh added `6` registry-backed `Wave 4 series` quantile/probabilistic candidates covering quantile and Gaussian uncertainty profiles across the `global` , `Fw` , and `Bw` surfaces.

Surface	Profile	Deterministic Curve	Candidate
`global`	`quantile_p10_p50_p90`	`p50`	`track2h_quantile_p10_p50_p90_global`
`Fw`	`quantile_p10_p50_p90`	`p50`	`track2h_quantile_p10_p50_p90_Fw`
`Bw`	`quantile_p10_p50_p90`	`p50`	`track2h_quantile_p10_p50_p90_Bw`
`global`	`gaussian_nll`	`mu`	`track2h_gaussian_nll_global`
`Fw`	`gaussian_nll`	`mu`	`track2h_gaussian_nll_Fw`
`Bw`	`gaussian_nll`	`mu`	`track2h_gaussian_nll_Bw`

The matrix source group is `track2h_quantile_probabilistic_registry` . The matrix now contains `147` candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
`Fw`	forward-only training or archive	forward TE Curve Verification Pipeline curves only
`Bw`	backward-only training or archive	backward TE Curve Verification Pipeline curves only
`global`	forward and backward training together	both directions, reported by direction and combined

The probabilistic heads are not compared as raw multi-channel tensors. The official matrix evaluates only the deterministic TE curve exposed by each checkpoint: `p50` for quantile heads and `mu` for Gaussian NLL heads.

Current Leaders

Overall Direction Leaders

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
`forward`	`rcim_retuned_GBM19_Fw`	0.001089	0.001299	2.372	4.912
`backward`	`periodic_gru_sequence_Bw`	0.002392	0.002639	5.466	14.820
`global forward`	`periodic_gru_sequence_global`	0.002777	0.003025	6.267	13.580
`global backward`	`periodic_gru_sequence_global`	0.002630	0.002872	6.010	12.693

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
`backward`	`tree_Bw`	0.003258	0.003651	7.051	14.116
`global forward`	`tree_global`	0.002998	0.003364	6.590	11.601
`global backward`	`tree_global`	0.003290	0.003702	7.118	13.703

Wave 4.2 Quantile Probabilistic Result

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
`global forward`	`track2h_gaussian_nll_global`	`gaussian_nll`	0.002951	0.003246	6.524	15.237
`Bw`	`track2h_quantile_p10_p50_p90_Bw`	`quantile_p10_p50_p90`	0.002935	0.003250	6.307	16.529
`Bw`	`track2h_gaussian_nll_Bw`	`gaussian_nll`	0.003001	0.003303	6.488	14.856
`global backward`	`track2h_gaussian_nll_global`	`gaussian_nll`	0.003068	0.003372	6.627	15.928
`Fw`	`track2h_gaussian_nll_`	`gaussian_nll`	0.003156	0.003415	7.008	10.991

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
	Fw					
global forward	track2h_quantile_p10_p50_p90_global	quantile_p10_p50_p90	0.003188	0.003469	7.059	12.765
Fw	track2h_quantile_p10_p50_p90_Fw	quantile_p10_p50_p90	0.003276	0.003545	7.279	12.393
global backward	track2h_quantile_p10_p50_p90_global	quantile_p10_p50_p90	0.003563	0.003909	7.816	15.378

The best probabilistic `global` candidate improves over the best robust-loss global candidate from `0.003401` to `0.003009` combined curve-verification MAE. The best probabilistic `Bw` candidate improves over the best robust-loss `Bw` candidate from `0.003078` to `0.002935` curve-verification MAE. The best robust-loss `Fw` candidate remains slightly better than the best probabilistic `Fw` candidate: `0.003134` versus `0.003156` curve-verification MAE.

The strongest probabilistic `Bw` candidate also beats the static `tree_Bw` baseline on curve-verification MAE, `0.002935` versus `0.003258`, but it remains behind the accepted `periodic_gru_sequence_Bw` branch.

Visual Evidence

The `2026-06-12` collage and overlay bundles were regenerated after the probabilistic refresh. They now include explicit `Wave 4.2` quantile probabilistic sections for `Fw`, `Bw`, and `global` collage evidence, plus explicit `Fw` and `Bw` probabilistic overlay sections.

The visual package supports the matrix decision: the probabilistic heads are a useful dispersion-aware ingredient, especially for `Bw` and the global surface, but they do not replace the strongest periodic temporal branch or the forward paper-reference leader.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-12	`Wave 4.2` quantile/probabilistic refresh	`6` quantile and Gaussian `global`, `Fw`, and `Bw` candidates	included in the `147`-candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-06-11	`Wave 4.1` robust-loss dispersion-aware refresh	`9` robust `global`, `Fw`, and `Bw` candidates	included in the `141`-candidate matrix	dated collage and overlay bundles regenerated with Wave 4 series sections	verified exploratory baseline; not promoted
2026-06-10	`Wave 3.3` curve-aware training refresh	`12` pointwise-control, centered-shape, offset, and full-composite `global`, `Fw`, and `Bw` candidates	included in the `132`-candidate matrix	dated collage and overlay bundles regenerated with Wave 3.3 sections	verified exploratory baseline; not promoted
2026-06-08	`Wave 3.2` harmonic-offset probe refresh	`6` clean and harmonic `global`, `Fw`, and `Bw` candidates plus `3` rechecked Wave 3.1 candidates	included in the `120`-candidate matrix	dated collage and overlay bundles regenerated with Wave 3.2 sections	verified exploratory baseline; not promoted
2026-06-04	`Wave 3.1` offset-aware probe refresh	`3` `global`, `Fw`, and `Bw`	included in the `114`-candidate matrix	dated collage and overlay bundles	verified exploratory baseline; not promoted

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
		sequential residual offset candidates		regenerated	
2026-05-28	`Wave 2.3` residual harmonic temporal hybrid refresh	`18` `global`, `Fw`, and `Bw` residual harmonic GRU/LSTM candidates	included in the `111`-candidate matrix	collage and overlay reports refreshed with `Wave 2.3` sections	verified exploratory baseline; not promoted over `Wave 2.2` or accepted TE Curve Verification Pipeline baselines
2026-05-26	`Wave 2.2` harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM `global`, `Fw`, and `Bw` candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch

Closeout Decision

`Wave 4.2` quantile/probabilistic regression does not change the accepted `TE Curve Verification Pipeline` baseline. The current direction-parallel decision remains:

- `Fw`: `rcim_retuned_GBM19_Fw` remains the strongest overall forward candidate; within this package, `track2h_gaussian_nll_Fw` is the strongest forward-only probabilistic candidate, but `track2h_gaussian_nll_global` performs better on forward curves.
- `Bw`: `periodic_gru_sequence_Bw` remains the strongest practical repository-owned backward candidate; `track2h_quantile_p10_p50_p90_Bw` is the strongest probabilistic backward candidate and improves over both the robust-loss Wave 4 series backward candidate and `tree_Bw` on curve-verification MAE.
- `global`: `periodic_gru_sequence_global` remains the strongest repository-owned bidirectional neural candidate; within the probabilistic package, `track2h_gaussian_nll_global` is clearly the best global surface and improves over the best robust-loss global Wave 4 series result. The next modeling step should keep probabilistic losses as a validated ingredient for dispersion-aware training. The evidence supports moving to mixture-density heads and latent-state or hysteresis-aware variants before deciding which mechanisms belong inside the integrated multi-task / multi-head branch.